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UNIVERSITEIT
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Master thesis submitted in partial fulfilment of the requirements for the degree of
Master of Science in Applied Sciences and Engineering: Computer Science

END-TO-END LEARNING TO PREDICT ARCHEOLOGICAL POTENTIAL

Integrating Deep Learning for Improved
Archeological Site Prediction

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August 2024

Promotors: Prof. Dr. Pieter Libin and Prof. Dr. Ralf Vandam (Faculty of Arts
and Philosophy)

Advisors: Simon Jaxy and Dr. Patrick Willelt (Faculty of Arts and Philosophy)
Science and Bio-Engineering Sciences



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END-TO-END LEARNING VOOR HET VOORSPELLEN VAN ARCHEOLOGISCH POTENTIEEL

Deep learning integratie voor verbeterde
voorspelling van potentiële
archeologische sites

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Abstract

Archaeologists uncover guiding factors in human history, providing deep knowledge about the evolution of societies through the study of artefacts found at archaeological sites. The research and documentation of these archaeological sites is often time-intensive and costly. To assist archaeologists in this work, predictive models are used to identify potential sites based on local geographical features. The models classically use feature extraction on which a regression analysis follows. To minimise the possible introduction of biases in feature extraction, an end-to-end method is proposed in this work. Similarly, naive Bayes is introduced as a possible tool for preliminary efficient analysis of archaeological potential while eliminating the need for manual parameter selection, further reducing possible biases. Both are trained on geospatial information gathered by remote sensing techniques and annotated features such as existing archaeological sites, distance to historical road networks and distance to urban centres. The methods were compared and evaluated against an existing technique in the Sagalassos survey area, situated in the western-Taurus mountain range in Anatolia. In the applied experimental setting, naive Bayes demonstrates to be superior against the end-to-end approach and existing method. Multiple complexities, introduced by the lack a general benchmark, are revealed in the analysis and prevent global claims. This work takes the first steps towards integration of much promising neural networks in the research field and can be used as a basis for further research.